Pizza Restaurant Bot (R)

Status	Done
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Classes	Live 06 - R Programming Foundation

Your tasks:

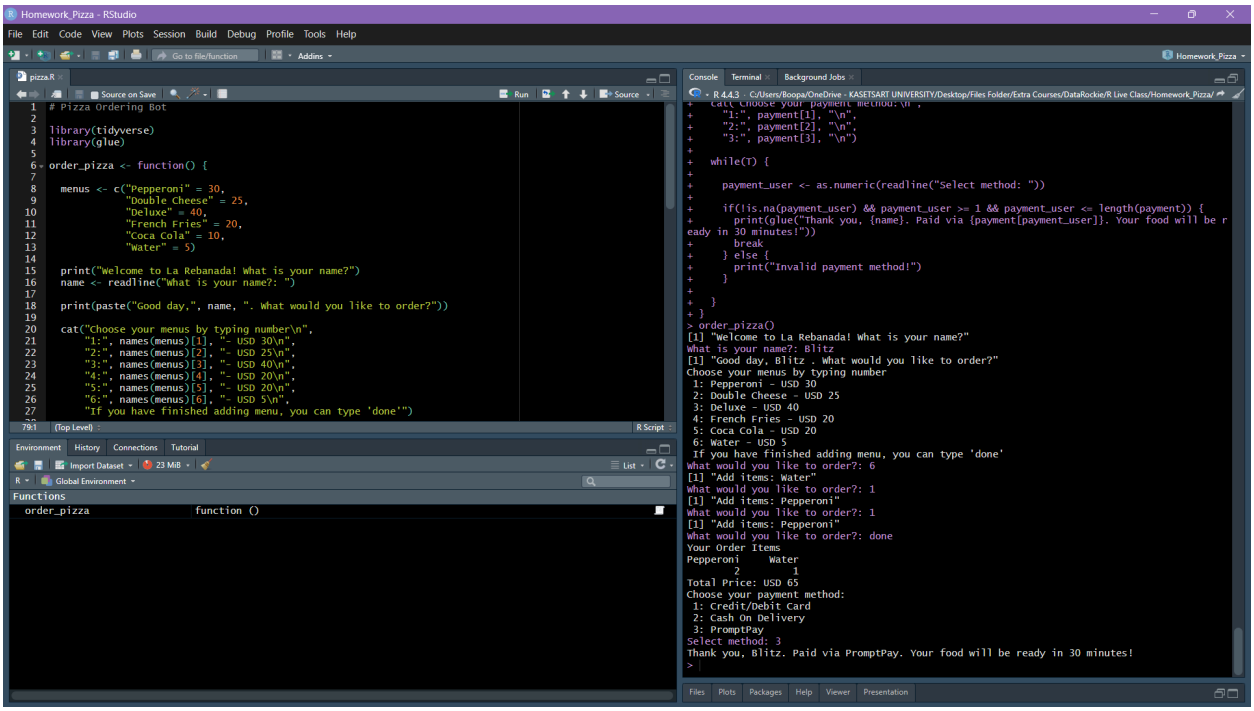
- imagine you open pizza restaurant and service by using application or bot
- you can create this freely. it's up to you

Here's the sample code, this might help:

```
# sample code
order_pizza <- function() {
  # bot: greeting
  print("Hi!")
  u-name <- readline("What's your name? ")
  print("Welcome to our online restaurant, ", u_name)
  # bot: there are 3 pizza menus
  # user: order pizza
  # bot summarize total order + cost
}

menus <- c(haiwaian = 20, cheese = 25, coke = 5)
menus[1]
menus[1] * 2
menus[3] * 3
```

My Code




```r
# Pizza Ordering Bot

library(tidyverse)
library(glue)

order_pizza <- function() {

  menus <- c("Pepperoni" = 30,
             "Double Cheese" = 25,
             "Deluxe" = 40,
             "French Fries" = 20,
             "Coca Cola" = 10,
             "Water" = 5)

  print("Welcome to La Rebanada! What is your name?")
  name <- readline("What is your name?: ")

  print(paste("Good day,", name, ". What would you like to order?"))
```

```
  cat("Choose your menus by typing number\n",
      "1:", names(menus)[1], "- USD 30\n",
      "2:", names(menus)[2], "- USD 25\n",
      "3:", names(menus)[3], "- USD 40\n",
      "4:", names(menus)[4], "- USD 20\n",
      "5:", names(menus)[5], "- USD 20\n",
      "6:", names(menus)[6], "- USD 5\n",
      "If you have finished adding menu, you can type 'don
e'")

  cart <- c()


  while (T) {
    order_menus <- readline("What would you like to order?:
")

    if (order_menus == "done") {
      break
    }

    choices <- suppressWarnings(as.numeric(order_menus))

    if(!is.na(choices) && choices >= 1 && choices <= length(m
enus)) {
      cart <- c(cart, choices)
      item_name <- names(menus)[choices]
      print(paste("Add items:", item_name))
    } else {
      print("Invalid menu! Please select available menus.")
    }
  }

  if (length(cart) == 0) {
    return("No items ordered. ")
```

```
  }

  ordered_items <-  names(menus)[cart]
  total_price <-  sum(menus[cart])

  print(table(ordered_items, dnn = "Your Order Items"))
  print(glue("Total Price: USD {total_price}"))

  payment <- c("Credit/Debit Card", "Cash On Delivery", "Prom
ptPay")
  cat("Choose your payment method:\n",
    "1:", payment[1], "\n",
    "2:", payment[2], "\n",
    "3:", payment[3], "\n")

  while(T) {

    payment_user <- as.numeric(readline("Select method: "))

    if(!is.na(payment_user) && payment_user >= 1 && payment_u
ser <= length(payment)) {
      print(glue("Thank you, {name}. Paid via {payment[paymen
t_user]}. Your food will be ready in 30 minutes!"))
      break
    } else {
      print("Invalid payment method!")
    }

  }
}
```

This is my pizza restaurant bot. Let’s break down into a step.

Steps

In this code, I installed `tidyverse` and `glue` function.

And the function created as `order_pizza()` with require no argument.

Start of with menus. I stored data including name to each value in vector.

```
menus <- c("Pepperoni" = 30,
           "Double Cheese" = 25,
           "Deluxe" = 40,
           "French Fries" = 20,
           "Coca Cola" = 10,
           "Water" = 5)
```

Which it's actually menu name and price.

Next, I use `readline()` to welcome a customer and also ask customer name for greeting.

```
print("Welcome to La Rebanada! What is your name?")
name <- readline("What is your name?: ")

print(paste("Good day,", name, ". What would you like to or
der?"))
```

Then I printed out menu list to allow customer to know menu name, menu number, and price.

```
cat("Choose your menus by typing number\n",
    "1:", names(menus)[1], "- USD 30\n",
    "2:", names(menus)[2], "- USD 25\n",
    "3:", names(menus)[3], "- USD 40\n",
    "4:", names(menus)[4], "- USD 20\n",
    "5:", names(menus)[5], "- USD 20\n",
    "6:", names(menus)[6], "- USD 5\n",
```

```
      "If you have finished adding menu, you can type 'don
e'")
```

To allow customer to add items in cart, I have created empty cart as empty vector.

```
  cart <- c()
```

Next, I started a loop control flow.

Receive menu number by customer input with `readline()`.

This is `while` loop where if you continue to order food, it will keep looping until you finish buying by typing `done`.

```
  while (T) {
    order_menus <- readline("What would you like to order?:
")

    if (order_menus == "done") {
      break
    }
```

Customer order will store as index in vector.

```
    choices <- suppressWarnings(as.numeric(order_menus))
```

I have to convert to NUMERIC because `readline()` always return as character.

And `suppressWarnings()` uses for suppress `as.numeric()` error when user types `done` because `done` is character and it will show warning message so I get rid of that since it works normally.

This condition is for checking is the menu number is valid or not.

If valid, it stores to `cart` and it shows status that food has been added to a cart by using `item_name`.

If not, it shows a warning message and you have to input a valid one.

```
    if(!is.na(choices) && choices >= 1 && choices <= length(m
enus)) {
      cart <- c(cart, choices)
      item_name <- names(menus)[choices]
      print(paste("Add items:", item_name))
    } else {
      print("Invalid menu! Please select available menus.")
    }
  }
```

Now, we are outside the loop.

If `cart` is empty, it will show message that it has nothing.

```
  if (length(cart) == 0) {
    return("No items ordered. ")
  }
```

And you need to order again by execute `order_pizza()` function.

Once order complete and with food in a cart, it returns summary of product items and total price.

```
  ordered_items <-  names(menus)[cart]
  total_price <-  sum(menus[cart])

  print(table(ordered_items, dnn = "Your Order Items"))
  print(glue("Total Price: USD {total_price}"))
```

Finally, user has to choose payment method.

There are 3 ways to choose.

```
  payment <- c("Credit/Debit Card", "Cash On Delivery", "Prom
ptPay")
  cat("Choose your payment method:\n",
    "1:", payment[1], "\n",
    "2:", payment[2], "\n",
    "3:", payment[3], "\n")
```

And then I created another loop.

If valid, it will print final message that order is completed.

If not, it shows warning message and you have to input a valid one.

```
  while(T) {

    payment_user <- as.numeric(readline("Select method: "))

    if(!is.na(payment_user) && payment_user >= 1 && payment_u
ser <= length(payment)) {
      print(glue("Thank you, {name}. Paid via {payment[paymen
t_user]}. Your food will be ready in 30 minutes!"))
      break
    } else {
      print("Invalid payment method!")
    }

  }
```

And that's it. Let's see some scenarios.

Result

Scenario 1: Completed Order

```
> order_pizza()
[1] "Welcome to La Rebanada! What is your name?"
What is your name?: Blitz
[1] "Good day, Blitz . What would you like to order?"
Choose your menus by typing number
 1: Pepperoni - USD 30
 2: Double Cheese - USD 25
 3: Deluxe - USD 40
 4: French Fries - USD 20
 5: Coca Cola - USD 20
 6: Water - USD 5
 If you have finished adding menu, you can type 'done'
What would you like to order?: 6
[1] "Add items: Water"
What would you like to order?: 1
[1] "Add items: Pepperoni"
What would you like to order?: 1
[1] "Add items: Pepperoni"
What would you like to order?: done
Your Order Items
Pepperoni     Water
        2         1
Total Price: USD 65
Choose your payment method:
 1: Credit/Debit Card
 2: Cash On Delivery
 3: PromptPay
Select method: 3
Thank you, Blitz. Paid via PromptPay. Your food will be ready
in 30 minutes!
```

Scenario 2: No Item in a Cart

```
> order_pizza()
[1] "Welcome to La Rebanada! What is your name?"
What is your name?: B
[1] "Good day, B . What would you like to order?"
Choose your menus by typing number
 1: Pepperoni - USD 30
 2: Double Cheese - USD 25
 3: Deluxe - USD 40
 4: French Fries - USD 20
 5: Coca Cola - USD 20
 6: Water - USD 5
 If you have finished adding menu, you can type 'done'
What would you like to order?: done
[1] "No items ordered. "
```

Scenario 3: Completed Order with Warning Message

```
> order_pizza()
[1] "Welcome to La Rebanada! What is your name?"
What is your name?: A
[1] "Good day, A . What would you like to order?"
Choose your menus by typing number
 1: Pepperoni - USD 30
 2: Double Cheese - USD 25
 3: Deluxe - USD 40
 4: French Fries - USD 20
 5: Coca Cola - USD 20
 6: Water - USD 5
 If you have finished adding menu, you can type 'done'
What would you like to order?: 7
[1] "Invalid menu! Please select available menus."
What would you like to order?: 1
[1] "Add items: Pepperoni"
What would you like to order?: 1
```

```
[1] "Add items: Pepperoni"
What would you like to order?: 2
[1] "Add items: Double Cheese"
What would you like to order?: 6
[1] "Add items: Water"
What would you like to order?: done
Your Order Items
Double Cheese     Pepperoni         Water
            1             2             1
Total Price: USD 90
Choose your payment method:
 1: Credit/Debit Card
 2: Cash On Delivery
 3: PromptPay
Select method: 5
[1] "Invalid payment method!"
Select method: 1
Thank you, A. Paid via Credit/Debit Card. Your food will be r
eady in 30 minutes!
```